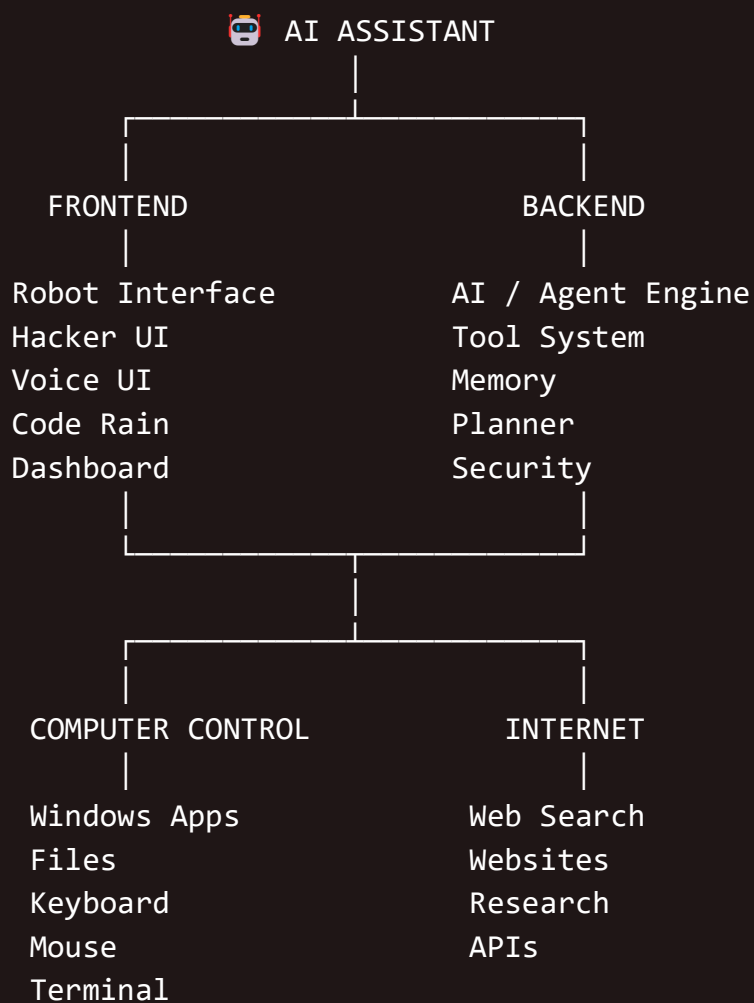




100-Day AI Assistant Roadmap

Final goal



PHASE 1 — 20 DAYS

Frontend + Robot/Hacker Interface

Goal: Build a beautiful interface first, even before the AI is connected.

Days 1–2 — Project setup

Create:

```
AI-ASSISTANT/  
├── frontend/  
├── backend/  
├── agent/  
├── tools/  
├── memory/  
├── database/  
├── tests/  
├── config/  
└── README.md
```

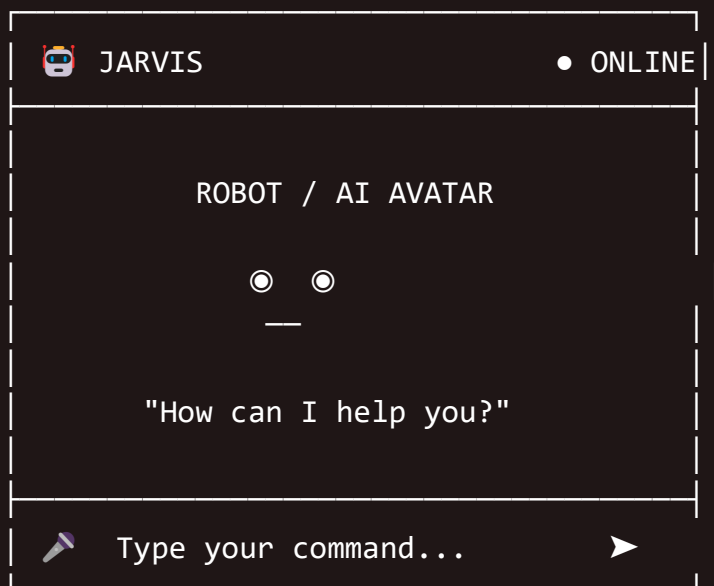
Set up:

- Git
- GitHub repository
- Python environment
- Node.js
- Frontend framework
- Basic backend

Deliverable: Project runs locally.

Days 3–5 — Main UI

Build:



Create:

- Header
- AI avatar
- Chat area
- Command box
- Microphone button
- Send button
- Status indicator

Days 6–8 — Hacker interface

Add:

- Dark theme
- Matrix/code rain
- Terminal panel
- Glowing elements
- Animated status indicators
- CPU/RAM display
- Network status
- Current time
- Task status

Example:

SYSTEM STATUS

CPU	24%
RAM	53%
NETWORK	ONLINE
AI	READY
TOOLS	12
TASKS	02

Days 9–11 — Robot animation

Create your robot/avatar.

States:

```
graph TD
  IDLE --> LISTENING
  LISTENING --> THINKING
  THINKING --> WORKING
  WORKING --> SUCCESS
  SUCCESS --> ERROR
```

The robot should visually change depending on what the agent is doing.

Days 12–14 — Chat interface

Implement:

```
graph TD
  USER --> Message_bubble[Message bubble]
```

```
graph TD
  AI --> Response_bubble[Response bubble]
```

Add:

- Markdown
- Code blocks
- Copy button

- Clear conversation
- Conversation history

Days 15–17 — Dashboard

Create panels for:

Tasks
Files
Applications
Memory
Tools
System
Settings

Example:

TASKS	CURRENT TASK
✓ Research • Download ○ Report	Researching... 70%

Days 18–19 — Voice interface

Build the UI for:

🎤 Listening...
↓
"Open Chrome"
↓
🤖 Thinking...
↓
Opening Chrome...

Initially the UI can use dummy responses.

Day 20 — Phase 1 test

By Day 20 you should have:

- ✓ Robot UI
- ✓ Hacker theme
- ✓ Code rain
- ✓ Chat interface
- ✓ Voice interface UI
- ✓ Dashboard
- ✓ Task panel

- ✓ Settings panel
- ✓ Animations

Phase 1 milestone: You have a beautiful AI assistant interface, but the brain isn't connected yet.

PHASE 2 — 40 DAYS

Backend + APIs + Computer Control

This is the biggest phase.

Days 21–25 — Backend foundation

Build FastAPI:

```
frontend
  ↓
FastAPI
  ↓
Agent Engine
```

Create APIs:

```
POST /chat
POST /command
GET /status
GET /tasks
POST /tasks
GET /memory
```

Days 26–30 — AI connection

Connect your LLM.

Architecture:

```
User
  ↓
Frontend
  ↓
FastAPI
  ↓
Agent
  ↓
```

LLM
↓
Response
↓
Frontend

First successful command:

"Hello"

Assistant:

"Hello! I'm ready."

Days 31–35 — Tool system

Create the most important part of the project.

```
tools/  
├─ computer.py  
├─ browser.py  
├─ files.py  
├─ terminal.py  
├─ system.py  
├─ communication.py  
└─ research.py
```

The AI should be able to choose tools.

Example:

User:
Open Chrome

AI:
Tool = open_application
Argument = Chrome

Tool executes

AI:
Chrome is now open.

Days 36–40 — Windows control

Build controlled tools for:

- Open application
- Close application
- Mouse movement
- Click
- Double-click
- Keyboard input
- Keyboard shortcuts
- Screenshots
- Window detection
- Window switching

Example:

"Open Notepad and type Hello"

↓

`open_app("Notepad")`

↓

`focus_window()`

↓

`type_text("Hello")`

↓

`verify_screen()`

Days 41–45 — File management

Implement:

Search files

Read files

Create files

Copy files

Move files

Rename files

Create folders

Open files

Supported initially:

- TXT
- PDF
- DOCX
- XLSX
- PPTX
- CSV
- Images

Example:

"Find my BAI402 PDF."

Agent:

Search

↓

Find file

↓

Show result

Days 46–50 — Browser automation

Build browser tools:

```
open_browser()
search_web()
open_url()
click()
type()
read_page()
download()
```

Then:

"Search for VTU BAI402 syllabus."

Agent:

Open browser

↓

Search

↓

Read results

↓
Open relevant source
↓
Extract information

Days 51–55 — Research engine

Build:

Research
↓
Search multiple sources
↓
Read sources
↓
Extract information
↓
Compare
↓
Summarize
↓
Cite sources

Example:

"Research the best cloud storage architecture for my project."

Days 56–60 — System tools

Add safe system operations:

Get CPU usage
Get RAM
Get storage
Get battery
Get network status
Lock computer
Restart
Shutdown
Open settings

For sensitive actions, implement confirmation:



You requested:

Shutdown computer

[Cancel] [Confirm]

Do **not** implement authentication bypass/unlocking mechanisms that circumvent Windows security.

Days 61–65 — Voice engine

Implement:

Microphone



Speech-to-text



AI



Action



Text-to-speech

Example:

"Jarvis, open YouTube."



Speech recognition



"open YouTube"



Agent



Browser tool



YouTube

Days 66–60? — Correction: Days 66–70

Task manager

Build long-running tasks.

Example:

"Research AI internships, create a comparison spreadsheet, and save it."

The system creates:

TASK #1024

- | | |
|------------------------|---|
| 1. Search internships | ✓ |
| 2. Collect information | ✓ |
| 3. Compare | ● |
| 4. Create spreadsheet | ○ |
| 5. Save file | ○ |

Add:

- Pause
- Resume
- Cancel
- Retry
- Progress

Day 70 — Phase 2 milestone

At this point your assistant should be able to:

CHAT

VOICE



AI



PLAN



TOOLS

- Windows
- Files
- Browser
- Terminal
- Research
- System

This is your **first real AI agent**.

PHASE 3 — 20 DAYS

Knowledge + Training + Intelligence

Now we make it smarter.

Important distinction:

Don't start by training an LLM from scratch.

For your assistant, use:

LLM
+
Prompting
+
RAG
+
Memory
+
Tools
+
Examples
+
Evaluation

Days 71–73 — Agent personality/instructions

Create your core system instructions.

Define:

Identity
Behavior
Rules
Tool usage
Safety
Communication style
Error handling

Example: You are a desktop AI assistant.

Your responsibilities:

1. Understand the user's goal.
2. Break complex requests into tasks.
3. Select appropriate tools.

4. Execute tasks sequentially.
5. Verify important actions.
6. Report what was completed.
7. Never claim an action succeeded without verification.

Days 74–77 — RAG knowledge system

Build:

Documents



Text extraction



Chunking



Embeddings



Vector database



Retriever



AI

Give it:

- Your project documentation
- Technical documentation
- Personal knowledge files
- Study material
- Manuals

Now:

"What is in my BAI402 notes?"

can be answered from your documents.

Days 78–80 — Long-term memory

Create:

memory/

├─ preferences

├─ projects

└─ tasks

└─ conversations
└─ facts

The agent can remember useful information across sessions.

Days 81–83 — Planning engine

Implement:

```
GOAL
↓
PLAN
↓
TASK 1
↓
OBSERVE
↓
TASK 2
↓
OBSERVE
↓
TASK 3
↓
VERIFY
```

This is one of the most important parts.

Days 84–86 — Self-correction

If a tool fails:

```
Tool failed
↓
Read error
↓
Diagnose
↓
Try alternative
↓
Verify
```

Example:

Chrome failed to open.

Agent:

- Check whether Chrome exists
- Try alternative launch method
- Verify window

Days 87–90 — Agent evaluation

Create a test suite.

Computer tests

Open Notepad
Open Chrome
Type text
Take screenshot
Create folder
Find file

AI tests

Simple question
Complex question
Multi-step task
Wrong command
Ambiguous command
Tool failure

Track:

Success rate
Tool accuracy
Task completion
Error rate
Response time

PHASE 4 — 20 DAYS



Debugging + Testing + Hosting

Days 91–93 — Full integration

Connect everything:

Frontend



API



Agent



LLM



Planner



Memory



Tools



Windows

Test complete workflows.

Days 94–96 — Security

Implement:

Permission system

SAFE

LOW RISK

HIGH RISK

Authentication

Your assistant should have its own login.

Secrets

Never put API keys directly inside frontend code.

Use:

`.env`

and secure secret storage.

Logging

Record:

Timestamp

User command

Tool used

Result

Error

Permission

But avoid logging passwords, tokens, or private message contents unnecessarily.

Days 97–98 — Performance

Optimize:

- API response time
- Memory
- Database
- AI requests
- Tool execution
- UI animations

Add caching where appropriate.

Day 99 — Final testing

Run a complete test.

Test 1

"Open Chrome."

Test 2

"Find my BAI402 notes."

Test 3

"Read Module 3 and create 20 MCQs."

Test 4

"Research this topic and create a report."

Test 5

"Open WhatsApp and prepare a message to Varun."

For sending messages, use a confirmation step before the final send.

Test 6

"Create a Python project."

Test 7

"Check my CPU and RAM."

Test 8

"Shutdown the computer."

Confirmation:

⚠ Shutdown requested.

Are you sure?

[YES] [CANCEL]

Day 100 — 🚀 Launch Day

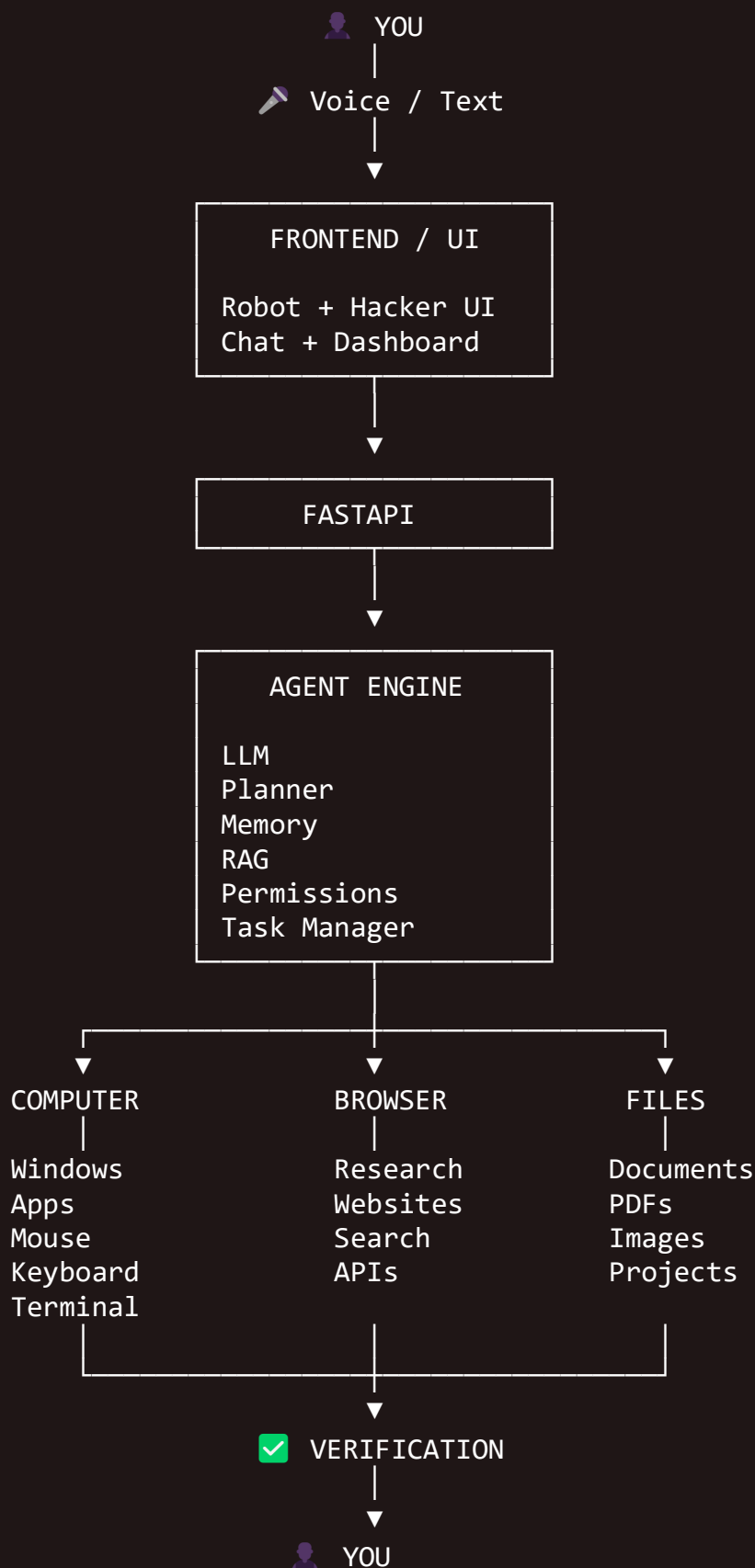
Final checklist:

🤖 AI ASSISTANT v1.0
✓ AI Brain
✓ Voice
✓ Robot UI
✓ Hacker Interface
✓ Code Rain
✓ Memory
✓ RAG
✓ Windows Control
✓ File Control
✓ Browser
✓ Research
✓ Terminal
✓ Task Manager
✓ Planner
✓ Error Recovery
✓ Permissions
✓ Logging
✓ Testing

📅 Your 100-day breakdown

Phase	Days	Main objective
🎨 Phase 1	1–20	Robot UI + Hacker interface
🧠 Phase 2	21–70	Backend + APIs + Tools + Computer control
🧠 Phase 3	71–90	Knowledge + Memory + Planning + Agent intelligence
🔧 Phase 4	91–100	Testing + Security + Debugging + Deployment

Final architecture



One important change I'd make to your original plan: don't wait until Phase 3 to give the assistant its "structures." Start building the **tool architecture and agent interface during Phase 2**, then Phase 3 makes those tools intelligent through planning, RAG, memory, and evaluation. That will make the 100-day target much more realistic.